

Auckland Excavated Soil and Landfill Transport

Evidence Summary and Long-Distance Scenario

Prepared from Stats NZ and Auckland Council sources | October 2026

Construction focus	Auckland recorded 17,264 new dwelling consents in the year ended August 2026, the highest absolute total of any New Zealand region.
Material focus	Auckland Council identifies excavated soil as one of the largest single material streams sent to landfill. About 2.09 million tonnes were excavated and disposed of in 2022.
Selected origin	Warkworth is a major planned growth area with about 1,000 ha of Future Urban zoned land and substantial future residential and employment development.
Selected destination	A consented cleanfill/managed-fill site is located at 489 Hunua Road, Drury (Hunua Quarry). Warkworth-to-Drury road distance is approximately 92 km.

Important interpretation: Building consents are used here as a transparent proxy for construction activity. They do not measure every civil or infrastructure project. Likewise, available public evidence supports soil as a very large excavation/disposal stream, but does not prove that “soil” is the single most common geological material excavated at every Auckland construction site.

1. Why Auckland is the project focus

Stats NZ building-consent data provides a consistent regional measure of approved construction activity. In the year ended August 2026, Auckland had 17,264 new dwelling consents, compared with 8,933 in Canterbury and 3,219 in Waikato. On this measure, Auckland had the highest absolute level of residential construction approvals in New Zealand.

This does not mean every category of construction is highest in Auckland; rather, it gives a defensible and reproducible basis for selecting Auckland as the study region.

Region	New dwellings consented (year ended Aug 2026)	Comparison
Auckland	17,264	Highest
Canterbury	8,933	~52% of Auckland
Waikato	3,219	~19% of Auckland

2. Why excavated soil is the target material

Auckland Council’s Waste Assessment 2023 identifies soil disposal as “one of the largest single material streams” going to landfills within and outside the Auckland region. The assessment estimates that 2.09 million tonnes of soil were excavated in 2022 and disposed of to Class 3 and 4 managed/controlled fills or Class 5 cleanfills. Of this, about 1,238,532 tonnes were disposed of at Auckland Council-consented cleanfills and managed fills, while about 190,731 tonnes left the region for North Waikato fill sites.

The Council’s Waste Minimisation and Management Plan 2024 gives the same scale in rounded form: approximately two million tonnes of soils from Auckland are excavated and disposed of to landfill each year, with most landfilled soils coming from land development.

“Every year, approximately two million tonnes of soils from the Auckland region are excavated and disposed of to landfill. Most landfilled soils are from land development...”

Project interpretation: Excavated soil is therefore a strong material focus because its annual disposal volume is very large and is explicitly linked by Auckland Council to land development. The public sources reviewed do not split the full 2.09 million tonnes into clay, silt, sand and rock, so this report does not claim that clay or any other sub-type is the dominant component.

3. Long-distance transport scenario

3.1 Selected construction area: Warkworth

Warkworth was selected as a plausible long-distance origin because Auckland Council identifies it as a key satellite town and major greenfield growth area. The Warkworth Structure Plan covers about 1,000 ha of Future Urban zoned land and anticipates substantial residential and employment growth over the coming decades.

3.2 Selected disposal site: 489 Hunua Road, Drury

Auckland Council’s list of consented cleanfill or non-cleanfill sites includes Fletcher Concrete and Infrastructure Limited at 489 Hunua Road, Drury. The site is recorded as a cleanfill and is associated with managed-fill placement and quarry rehabilitation. This makes it a defensible disposal destination for a clean/managed-fill transport scenario.

3.3 Road distance

A public road-distance service reports approximately 92 km by road between Warkworth and Drury. Because this distance is town-to-town rather than a GPS route from a specific construction parcel to the gate at 489 Hunua Road, it should be treated as an approximate long-distance scenario rather than an Auckland-wide average.

Origin	Warkworth, Auckland
Destination	489 Hunua Road, Drury / Hunua Quarry cleanfill

Approx. one-way road distance	92 km
Approx. round trip	184 km
Scenario type	Long-distance / upper-distance illustration

4. Illustrative transport implication

To show why transport distance matters, an illustrative calculation can be made using the rounded annual soil quantity of 2.0 million tonnes and an assumed truck payload of 20 tonnes. This is a scenario calculation only; it does not mean all Auckland soil is transported from Warkworth to Drury.

Annual soil quantity used	2,000,000 t (rounded Council figure)
Assumed payload	20 t per truck load
Truck loads	100,000 loads
Loaded travel	$100,000 \times 92 \text{ km} = 9.2 \text{ million truck-km}$
Round-trip vehicle travel	$100,000 \times 184 \text{ km} = 18.4 \text{ million vehicle-km}$

Do not present 18.4 million vehicle-km as an observed Auckland total. It is an upper-distance illustration created from the selected Warkworth-Drury route and the 20 t truck assumption. A robust regional estimate would require the actual origin-destination distribution of soil movements and real truck payload data.

5. Conclusion

The evidence supports Auckland as a sensible study region because it has the highest absolute number of new dwelling consents in New Zealand on the latest Stats NZ regional data. Auckland Council also identifies excavated soil as one of the region's largest material streams sent to disposal facilities, with about 2.09 million tonnes excavated and disposed of in 2022 and the 2024 Waste Plan summarising the scale as approximately two million tonnes each year. For a deliberately long-distance transport scenario, Warkworth is a defensible construction-growth origin and the consented cleanfill at 489 Hunua Road, Drury is a defensible destination, producing an approximate Warkworth-to-Drury road distance of 92 km.

The report should therefore frame the result as: Auckland is the selected construction region; excavated soil is the selected high-volume material stream; and Warkworth-to-Drury is an illustrative long-distance disposal route. This wording keeps the conclusions aligned with what the sources actually demonstrate.

References

1. Stats NZ (2026), Building consents issued: August 2026.

<https://www.stats.govt.nz/information-releases/building-consents-issued-August-2026/>
Regional new dwelling consents; Auckland 17,264 in the year ended August 2026.

2. Auckland Council (2023), Waste Assessment 2023.

<https://www.aucklandcouncil.govt.nz/plans-projects-policies-reports-bylaws/our-plans-strategies/topic-based-plans-strategies/environmental-plans-strategies/docswastemanagementplan/waste-assessment-2023.pdf>

Section 4.3.1 Soils: 2.09 million tonnes excavated in 2022; disposal to Class 3-5 facilities; quantities disposed in Auckland and North Waikato.

3. Auckland Council (2024), Auckland's Waste Minimisation and Management Plan 2024.

<https://www.aucklandcouncil.govt.nz/plans-projects-policies-reports-bylaws/our-plans-strategies/topic-based-plans-strategies/environmental-plans-strategies/Documents/waste-minimisation-management-plan-towards-zero-waste.pdf>
“Approximately two million tonnes” of soils excavated and disposed to landfill each year; most from land development.

4. Auckland Council, Warkworth Structure Plan / growth information.

<https://ourauckland.aucklandcouncil.govt.nz/news/2019/06/green-light-for-warkworth-plan/>
Warkworth as a key satellite town; about 1,000 ha of greenfield/Future Urban land; substantial planned growth.

5. Auckland Council, Consented Cleanfill or Non Cleanfill Sites (as of 30 June 2025).

<https://www.aucklandcouncil.govt.nz/building-and-consents/resource-consents/types-resource-consents/earthworks/Documents/consented-landfills.pdf>
Lists Fletcher Concrete and Infrastructure Limited, 489 Hunua Road, Drury, as a cleanfill/managed-fill site.

6. DistancesFrom, Warkworth to Drury road distance.

<https://www.distancesfrom.com/distance-from-Warkworth-to-Drury-New-Zealand/DistanceHistory/29633985.aspx>
Reports approximately 92 km road distance between Warkworth and Drury. Used only for the illustrative route distance.

Source note: The supplied one-page Auckland Council excerpt used in this project reproduces the key 2024 Waste Plan statement that approximately two million tonnes of Auckland soils are excavated and disposed of to landfill each year, with most landfilled soils from land development.